



GuardianGrid

AI-Powered Multi-Agent Smart City Assistant

“One City. Multiple AI Guardians. Infinite Possibilities.”

Submission Details

01

SOURCE CODE REPOSITORY

<https://github.com/HarshitaChokhani/GuardianGrid>

DEMO VIDEO (Google Drive)

https://drive.google.com/drive/folders/1hF-nDw6hEbHkdnfdLhDOIEm99LrbkxuX?usp=drive_link

PROJECT DOCUMENTATION

Full project proposal — see pages 2 & 3 of this PDF

BUILT WITH

HTML · CSS · JavaScript — no build step, opens offline via index.html

Submitted by **Harshita Chokhani**

What the Prototype Delivers

02

- Citizen grievance portal — photo + text report, AI auto-classification, India-format complaint IDs & live tracking.
- Explainable AI Resolution Engine — 6 agents reason on each report: “6 autonomous decisions · 0 human actions.”
- Government / Officer console — complaints land auto-routed; resolving one updates the citizen instantly (loop closed).
- Nationwide routing — all 28 states + 8 UTs, ~600 cities, real municipal bodies (BMC, BBMP, MCD, GHMC, ...).



GuardianGrid

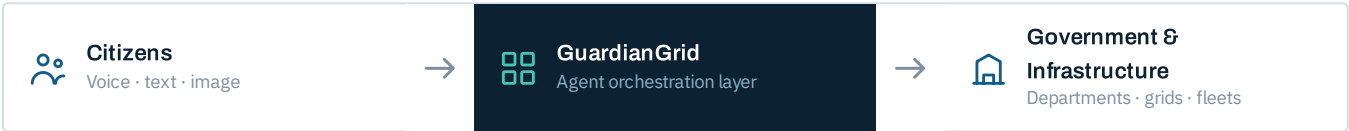
AI-Powered Multi-Agent Smart City Assistant

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Project Overview

01

GuardianGrid is a next-generation **multi-agent AI platform** that reinvents how citizens interact with city services. Rather than juggling separate portals and helplines, a citizen makes one request and a team of autonomous agents **collaborate to resolve civic issues, optimise operations, and improve engagement** in real time. In effect, it is an intelligent operating layer sitting between the people of a city and everything that runs it.



Problem Statement

02

Modern cities still run on fragmented systems that slow everything down:

- Citizens navigate dozens of disconnected government portals.
 - Departments operate in isolated, siloed systems.
 - Utility failures are reported and triaged manually.
 - Authorities lack predictive insight for proactive planning.
- Complaint resolution is slow and lacks transparency.
 - Congestion lengthens critical emergency response times.
 - Citizens receive delayed, inconsistent updates.
 - Rapid urban growth demands far smarter governance.

68%

of the world's population is projected to live in cities by 2050 (UN). Single-channel, manual service models simply cannot scale to that load — coordination has to become autonomous.

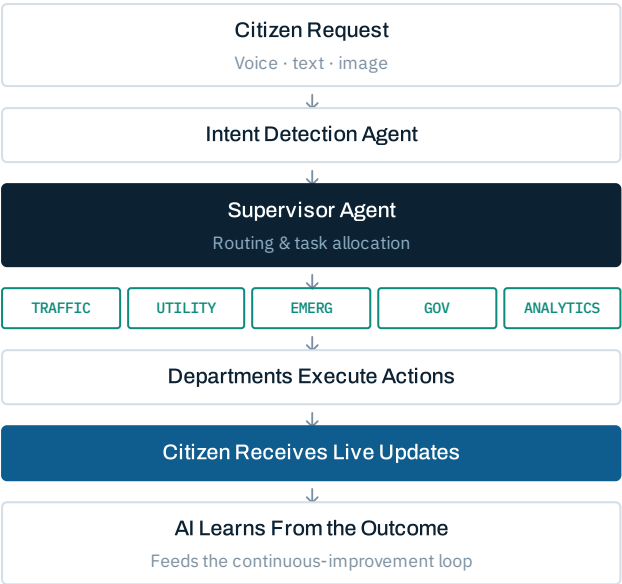
Proposed Solution

03

Not one chatbot, but a **collaborative ecosystem of specialised agents** that share context and make coordinated decisions on their own.

 Traffic Intelligence Predicts congestion, suggests alternate routes, and clears the way for emergency vehicles.	 Civic Services Handles garbage collection, road damage, streetlight failures and water-leakage complaints.	 Utility Monitoring Detects infrastructure failures, tracks power and water, and predicts maintenance needs.
 Emergency Response Coordinates ambulance, police and fire, finds the nearest responders, and streams live updates.	 Government Services Guides citizens through certificates, licences, taxes, schemes and appointment booking.	 City Analytics Builds dashboards, surfaces recurring issues, and forecasts future infrastructure problems.

Shared memory & orchestration — every agent communicates continuously over one shared context store, coordinating decisions in real time instead of working in isolation.



- 01 **One request, any format.** Voice, text or a photo of the issue enters through a single front door.
- 02 **Intent is understood** and structured before anything is routed anywhere.
- 03 **The Supervisor orchestrates**, allocating the task to the right specialised agent or agents.
- 04 **Agents act in parallel**, sharing context so the response stays coordinated, never duplicated.
- 05 **Departments execute** and the citizen is kept informed with live status throughout.
- 06 **Every outcome trains the system**, closing the loop and sharpening the next response.

Expected Impact

Challenge	GuardianGrid solution	Expected outcome
Traffic delays	AI rerouting & signal priority	Faster emergency response
Complaint tracking	Automated routing & live status	Reduced resolution time
Government services	Unified AI assistant	Better citizen experience
Infrastructure failures	Predictive monitoring	Lower maintenance cost
Emergency coordination	Multi-agent collaboration	Faster rescue operations

Innovation Highlights

- + Multi-agent AI architecture
- + Predictive analytics
- + Shared memory between agents
- + Autonomous decision making
- + Human-in-the-loop governance
- + Scalable cloud architecture
- + AI-to-AI collaboration
- + Real-time civic intelligence
- + Citizen-centric design

Future Scope

NEAR TERM	MID TERM	LONG TERM
IoT sensor integration CCTV analytics Smart public transportation	Digital twin cities Drone-based monitoring Disaster response coordination	AI-powered urban planning Sustainability monitoring Carbon footprint tracking

Conclusion

GuardianGrid is not merely a chatbot. It is an intelligent network of autonomous AI agents that collaborate to make cities more **responsive, efficient, transparent and citizen-centric** — turning fragmented services into one coordinated intelligence that anticipates rather than reacts.

"GuardianGrid envisions a future where every city is empowered by **collaborative AI agents** that anticipate challenges, coordinate solutions, and create safer, smarter, and more connected communities."